

```
<text x="5" y="15" fill="white" font-size="8pt" font-  
family="Verdana">we saw giant otters!</text>  
<rect x="0" y="0" width="120" height="90" stroke-  
width="1" stroke="black" fill="none"/>  
</svg>
```

## High Level Contrast of Canvas and SVG

The following is a high-level contrast of Canvas and SVG meant to frame a discussion of when to use one particular vector graphic technology over the other.

| A COMPARISON OF CANVAS AND SVG                                                                                                     |                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Canvas                                                                                                                             | SVG                                                                                                              |
| Pixel-based (canvas is essentially an image element with a drawing API)                                                            | Object Model-based (SVG elements are similar to HTML elements)                                                   |
| Single HTML element similar to <img> in behavior                                                                                   | Multiple graphical elements which become part of the Document Object Model (DOM)                                 |
| Visual presentation created and modified programmatically through script                                                           | Visual presentation created with markup and modified by CSS or programmatically through script                   |
| Event model/user interaction is coarse—at the canvas element only; interactions must be manually programmed from mouse coordinates | Event model/user interaction is object-based at the level of primitive graphic elements—lines, rectangles, paths |
| API does not support accessibility; markup-based techniques must be used in addition to canvas                                     | SVG markup and object model directly supports accessibility                                                      |

SVG is known as a retained mode graphics model persisting in an in-memory model. Analogous to HTML, SVG builds an object model of elements, attributes, and styles. When the <svg> element appears in an HTML5 document, it behaves like an inline block and is part of the HTML document tree.

Canvas is a bitmap with an immediate mode graphics application programming interface (API) for drawing on it. Canvas is a “fire and forget” model that renders its graphics directly to its bitmap and then subsequently has no sense of the shapes that were drawn; only the resulting bitmap stays around.

One way to think of these is that Canvas resembles the Windows GDI API, where you programmatically draw graphics to a window, and SVG